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CROPBOT
FARMER'S HELP

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Problem Statement

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Lack of accessible, localized, and easy-to-understand agricultural guidance causes poor farming decisions and financial losses for farmers.

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Team Structure

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Role	Team Member	Tech Stack	Module's Output
Interface Developer	Akhil Kotnala	React/ Vanilla JS/ CSS	The user-facing web app
Knowledge Architect	Shivaansh Gusain	Docling, Neo4j, Cypher, Docker	The populated, query-ready graph database
Agentic Engineer	Sandeep Jha	LangChain, LangGraph, Python, LLM Prompting	The reasoning engine and prompt logic
Integration Lead	Suryansh Bhatnagar	FastAPI, REST APIs, External APIs, Python	The routing server and external data tools

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The Solution

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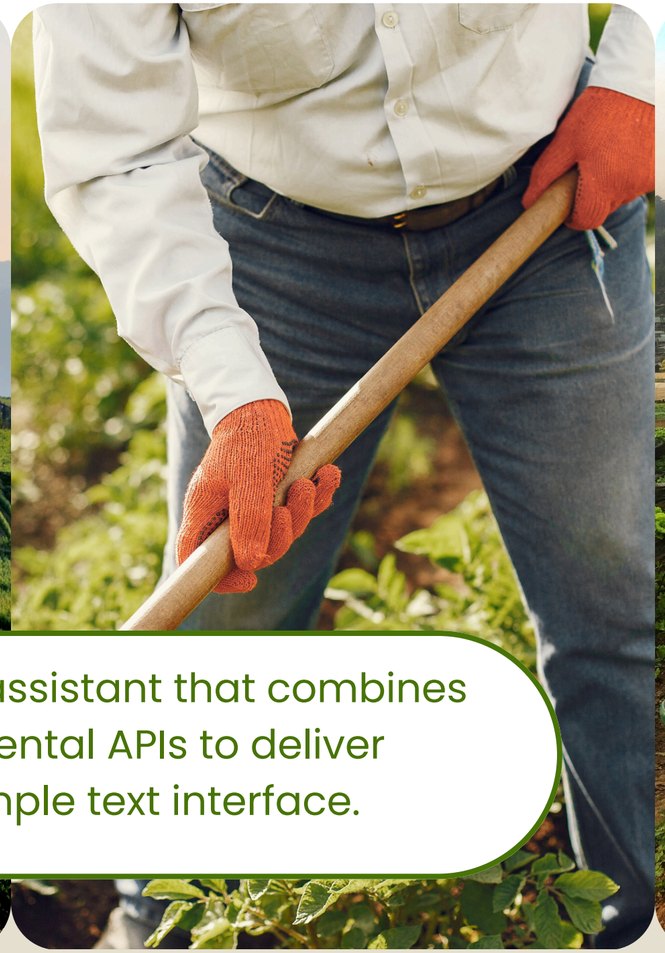

CropBot is a low-overhead, web-accessible AI agronomy assistant that combines a verified agricultural Knowledge Graph with live environmental APIs to deliver precise, hallucination-free farming guidance through a simple text interface.

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Why Now?

The Convergence of Rural Connectivity, AI Familiarity, and Government Infrastructure

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The Convergence

1. The Rural Digital Boom and Ultra-Low Data Costs

2. Mainstream AI Readiness

3. The Government's ₹2,817 Crore Digital Agriculture Mission (DAM)

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Key Features

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Real-Time Agricultural Intelligence

Integrates weather, soil, and location APIs to provide context-aware farming insights.

Comprehensive & Verified Knowledge

ICAR, Agrimly, Agritech & university sources covering crops, diseases, pests and cultivation, with periodic updates.

Knowledge Graph Integration

Connects agricultural entities and their relationships to enable relevant and contextual information retrieval.

Natural Language Understanding

LangChain and LLMs interpret farmer queries and intent, delivering simple, localized guidance.

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Product Roadmap

MVP

Our current model will be dealing with around 25 or more crops, multiple locations, web interface. It will be multi-lingual (Hindi, English), carefully look at the diseases, pests, seasonal alerts for recognizing climate. We will be looking forward to scale the model to a more large level and implement the model in an application interface.

Business Model

We operate an institutional B2B2FPO and B2G revenue model to keep the app free for farmers. Income is generated through annual SaaS subscriptions from Farmer Producer Organizations and state agriculture bodies looking to scale advisory reach.

Market Opportunity & Target Users

Our primary users are marginal farmers (<2 hectares) facing severe climate risks, alongside village FPO leads using the tools as an agronomy co-pilot. By partnering with 250+ regional FPO’s accross India, we aim to serve 500,000 active farmers within 3 years.

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Innovation and Uniqueness

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Interaction

One-on-one text AI — no browsing, no scrolling.

Intelligence

Real-time data, knowledge graph, and LLM reasoning working together.

Response

Instant, three-step actionable guidance — not long-form content.

Personalization

Aware of crop, location, and weather for every recommendation.

Core Purpose

Built to help farmers decide, not to sell them anything.

Bias

No selling pressure — advice stays mission-driven, not brand-driven.

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Competitive Differentiation

Bypassing data-heavy video and e-commerce bias to deliver instant, text-light agronomic truth.

Feature	CropBot (Our MVP)	Digital Green / Community	Agri-Input Apps (AgroStar/DeHaat)
How you use it	Simple 1-on-1 text chat	Watching videos & peer chat	Browsing a shop catalog
Type of Answer	Short 3-line steps	Group stories and long videos	Product links & price tags
Wait Time	Instant (LLM Logic)	Time to watch/download	Variable (Often sales talk)
Main Goal	Helping you decide	Learning from others	Selling you products
Selling Pressure	None (Unbiased)	None (Mission-driven)	High (Promoting brands)

Rollout ROAD MAP

Phase 1: Alpha— Knowledge Ingestion & Prototype

Build a rudimentary text-only Python terminal script to verify that the RAG pipeline retrieves the correct disease treatments without hallucinating.

Phase 2: Beta/MVP — Web Interface & API Integration

Develop a basic, responsive web app interface (text-in / text-out) supporting Hindi and English.

Phase 3: Pilot (Months 5-6) — Partner Rollout & Field Testing

Partner with a local Krishi Vigyan Kendra (KVK) or Farmer Producer Organization (FPO) in Uttarakhand to distribute the web app link to a closed group of 100-200 farmers.

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Thank You!

https://crop-bot-agro-based.vercel.app/